

- 7 (a) ${}^3_2\text{He} + {}^3_2\text{He} \rightarrow {}^4_2\text{He} + 2 {}^1_1\text{p} + Q$
 A numbers correct (4 and 1) B1
 Z numbers correct (2 and 1) B1 [2]
- (b) both nuclei have 2 protons B1
 the two isotopes have 1 neutron and two neutrons B1 [2]
 [allow 1 for 'same number of protons but different number of neutrons']
- (c) proton number and neutron number B1
 energy – mass B1
 momentum B1 [2]
- (d) (i) γ radiation B1 [1]
 (ii) product(s) must have kinetic energy B1 [1]
- (e) $13.8 \text{ MeV} = 13.8 \times 1.6 \times 10^{-19} \times 10^6 (= 2.208 \times 10^{-12})$ C1
 $60 = n \times 13.8 \times 1.6 \times 10^{-13}$
 $n = 2.7(2) \times 10^{13} \text{ s}^{-1}$ A1 [2]